Power Budget

Team Number:	101
Project Name:	**Automatic Pet Feeder**
Team Member Names:	**Caleb Ross, Connor Erwin, Gary Hoffland, Daksh Raghav Preetha**

All Major Components	**Component Name**	**Part Number**	**olyVoltageRan**	**#**	**MaximumCurre**	**talCurrent(mA)**	**Unit**
	Optical Sensor	TCST1103	1.2V to 70V	1	100	100	mA
	LED	151051RS11000	2.1V to 2.5V	2	100	200	mA
	Pushbutton	B3F-4055	0V to 24V	1	50	50	mA
	Opamp	MCP6004-I/P	1.8V to 6V	1	92	92	mA
	PIC Microcontroller Board	DM164150	1.8V to 5.5V	1	500	500	mA
	5V regulator	L7805CV	7.5V to 35V	1	1500	1500	mA
+5V Power Rail	**Component Name**	**Part Number**	**olyVoltageRan**	**#**	**MaximumCurre**	**talCurrent(mA)**	**Unit**
	Opamp	MCP6004-I/P	1.8V to 6V	1	92	92	mA
	Optical Sensor	TCST1103	1.2V to 70V	1	100	100	mA
	LED	151051RS11000	2V	2	100	200	mA
	Pushbutton	B3F-4055	0V to 24V	1	50	50	mA
	PIC Microcontroller Board	DM164150	1.8V to 5.5V	1	500	500	mA
					Subtotal	942	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	1177.5	mA
c2. Regulator or Source Choice	+5V Regulator	LM7805CV	7.5V to 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	322.5	mA
External Power Source 1	**Component Name**	**Part Number**	**olyVoltageRan**	**Output**	**MaximumCurre**	**talCurrent(mA)**	**Unit**
Power Source 1 Selection	Battery	NIMH-9V250-1B	7.0V to 9V	-8.4V	250	500	mA
Power Rails Connected to External Power Source 1	5V Regulator	L7805CV	7.5V to 35V	1	44.03	44.03	mA
					Total Remaining Current Available on External Power Source 2	455.97	mA

Component Name	**Part Number**	**Supply**	**Capacity**	**Required**	
Battery		+9V	500	44.03	
			Battery Life	11.35589371	hours